Internship Report — Day 10

Internship Role: Cloud Engineer Trainee Intern
Organization: Spektra Systems
Day: 10
Task Title: Deployment of Secure Reverse Proxy Architecture using Azure ARM Template

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1. Objective of the Task

The objective of today's task was to design and deploy a **secure multi-tier cloud architecture** using an **Azure ARM (Azure Resource Manager) template**. The deployment involved automating infrastructure provisioning through Infrastructure as Code (IaC) principles and understanding dependency management between Azure resources.

The architecture aimed to implement a **Reverse Proxy setup** where:

- A public-facing virtual machine acts as an **Nginx Proxy Server**
- A backend application virtual machine remains **hidden in a private subnet**
- Network security controls restrict direct public access to internal resources

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2. Technologies and Services Used

- Microsoft Azure ARM Templates
- Azure Virtual Machines (Linux – Ubuntu 22.04)
- Azure Virtual Network (VNet)
- Subnets (Public and Private)

- Network Security Groups (NSG)
- Public IP Address
- Azure VM Extensions (Custom Script Extension)
- Nginx Web Server

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3. Architecture Overview

The deployed architecture consisted of:

Network Layer

- A Virtual Network named **LinuxProxyVNet**
- Two subnets:
    - **Proxy-Subnet (Public Subnet)** – Hosts the proxy VM
    - **App-Subnet (Private Subnet)** – Hosts the internal application VM

Security Layer

- Network Security Group allowing:
    - SSH (Port 22)
    - HTTP (Port 80)
- Backend VM had **no public IP**, ensuring isolation.

Compute Layer

Two Linux Virtual Machines were deployed:

1. **Nginx Proxy VM**
    - Publicly accessible
    - Receives external traffic

    - Forwards requests internally

2. **Hidden App VM**
    - Private IP only (10.0.2.4)
    - Hosts internal web application
    - Accessible only via proxy server

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4. Implementation Process

Step 1: ARM Template Creation

A JSON ARM template was created containing:

- Parameters for username, password, VM size, and location
- Variables defining resource names and network configuration
- Resources defining networking and compute infrastructure

Step 2: Dependency Configuration

The `dependsOn` property was implemented to ensure correct deployment order:

- NSG created before Virtual Network
- VNet created before Network Interfaces
- NICs created before Virtual Machines
- Application VM configured before Proxy VM setup

This ensured Azure deployed resources sequentially without failures.

Step 3: Networking Deployment

- NSG rules were applied to both subnets.
- Public IP assigned only to proxy VM.

- Static private IP assigned to application VM.

Step 4: Virtual Machine Deployment

Ubuntu Linux VMs were automatically provisioned using ARM template image references.

Step 5: Automated Server Configuration

Azure Custom Script Extensions executed commands automatically:

Application VM Configuration

- Installed Nginx

Created a web page:

```
Secret App Server - Hidden in Private Subnet
```

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Proxy VM Configuration

- Installed Nginx
- Configured reverse proxy
- Forwarded traffic to private server (10.0.2.4)

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5. Deployment Execution

The deployment was performed using PowerShell:

```
New-AzResourceGroupDeployment `
  -ResourceGroupName "syedfuzail2009-rg" `
  -TemplateFile ".\vvvmmmm.json" `
  -Verbose
```

The verbose mode allowed observation of:

- Resource creation sequence

- Dependency execution
- Provisioning status

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6. Key Observations

- Azure deploys resources based on dependency mapping rather than template order.
- Proper `dependsOn` configuration prevents deployment conflicts.
- Private subnet architecture improves security by eliminating direct internet exposure.
- VM Extensions enable automatic configuration without manual SSH login.
- Reverse proxy architecture acts as a secure gateway between internet users and backend services.

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7. Learning Outcomes

Through this task, I learned:

- Practical implementation of **Infrastructure as Code (IaC)** using ARM templates.
- Importance of resource dependency management.
- Designing secure cloud network architectures.
- Reverse proxy implementation using Nginx.
- Automating server configuration using Azure VM Extensions.
- Secure separation of public and private workloads in Azure networking.

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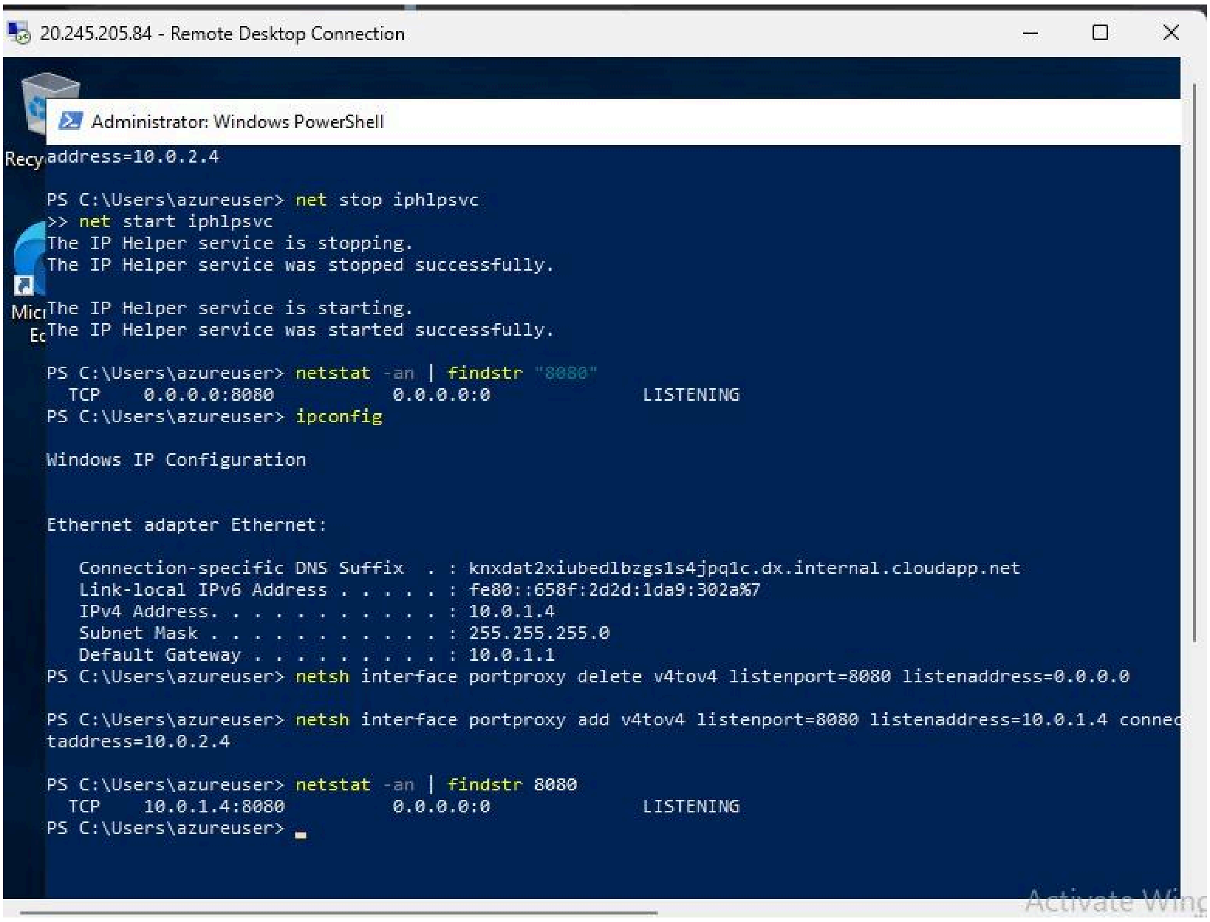
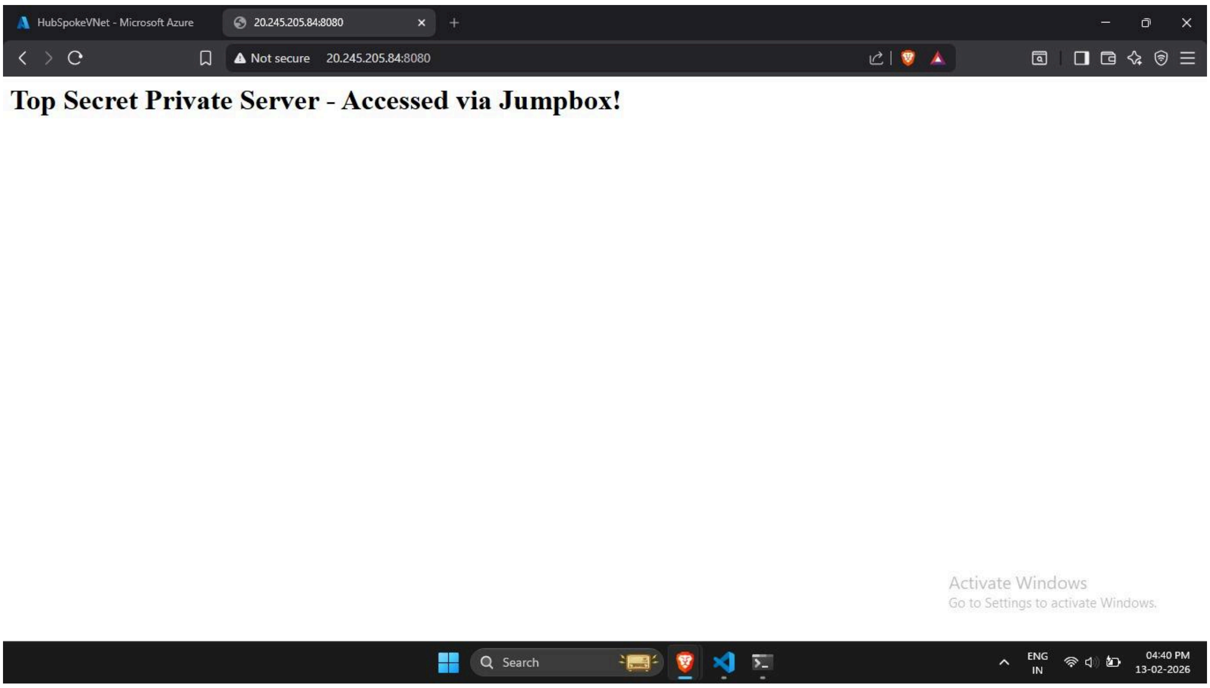
8. Conclusion

Day 10 focused on deploying a production-style secure architecture using ARM templates. The task demonstrated how Azure automation can provision complete infrastructure, configure servers, and enforce security controls without manual intervention. The reverse proxy model provided practical exposure to real-world cloud security and networking concepts commonly used in enterprise environments


20.245.205.84 - Remote Desktop Connection
Administrator: Windows PowerShell
address=10.0.2.4
PS C:\Users\azureuser> net stop iphlpsvc
>> net start iphlpsvc
The IP Helper service is stopping.
The IP Helper service was stopped successfully.
The IP Helper service is starting.
The IP Helper service was started successfully.
PS C:\Users\azureuser> netstat -an | findstr "8080"
TCP 0.0.0.0:8080 0.0.0.0:0 LISTENING
PS C:\Users\azureuser> ipconfig
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : knxdat2xiubedlbzgs1s4jpq1c.dx.internal.cloudapp.net
Link-local IPv6 Address . . . . . : fe80::658f:2d2d:1da9:302a%7
IPv4 Address. . . . . : 10.0.1.4
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . : 10.0.1.1
PS C:\Users\azureuser> netsh interface portproxy delete v4tov4 listenport=8080 listenaddress=0.0.0.0
PS C:\Users\azureuser> netsh interface portproxy add v4tov4 listenport=8080 listenaddress=10.0.1.4 connec
taddress=10.0.2.4
PS C:\Users\azureuser> netstat -an | findstr 8080
TCP 10.0.1.4:8080 0.0.0.0:0 LISTENING
PS C:\Users\azureuser>


HubSpokeVNet - Microsoft Azure
20.245.205.84:8080
Not secure 20.245.205.84:8080
Top Secret Private Server - Accessed via Jumpbox!
Activate Windows
Go to Settings to activate Windows.
Search
ENG
IN
04:40 PM
13-02-2026

Home

HubSpokeVNet

Virtual network

Diagnose issues with this virtual network | +2

Move | Delete | Refresh | Give feedback

Subscription (move)

Vanguard Class Training

Subscription ID

4dd60a42-8e1f-4d8b-b07a-fae39d4a2973

DNS servers

Azure provided DNS service

BGP community string

Configure

Virtual network ID

4f306e53-4597-4102-ac39-34b72f25f0da

Tags (edit)

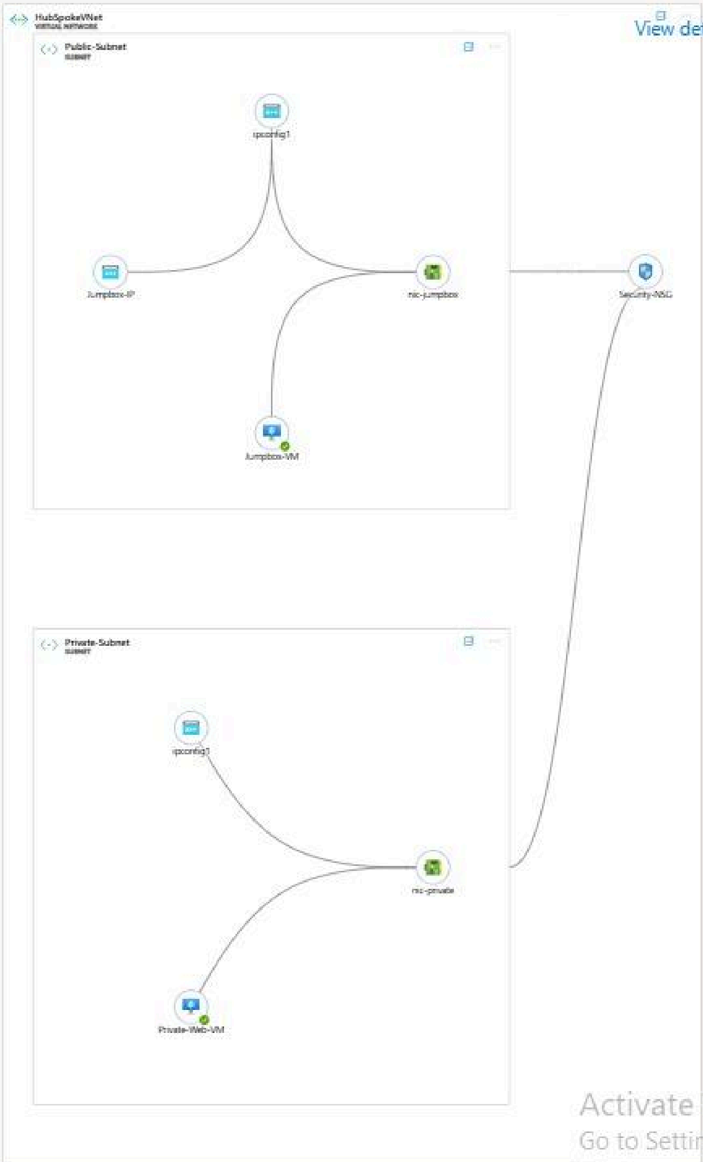
Add tags

Topology | Properties | Capabilities (5) | Recommendations | Tutorials

Virtual network

HubSpokeVNet

View detailed topology



Activate Windows
Go to Settings to activate Windows.